📘 FINAL PROJECT REPORT

Automating Data Import and Relationship Mapping Using Import Sets & Dot Walking

1. Abstract

This project focuses on automating the process of importing large datasets into the system and establishing relationships between tables using Import Sets and Dot Walking. Manual data entry is time-consuming and error-prone. By leveraging automated import mechanisms and relational mapping techniques, this project improves efficiency, accuracy, and scalability in data handling.

2. Introduction

Organizations deal with large volumes of data that need to be imported, transformed, and linked across multiple tables. Traditional methods require manual intervention, which increases the risk of inconsistencies.

This project provides a solution using:

- Import Sets for bulk data import
- Transform Maps for data conversion
- Dot Walking for relationship mapping between tables

3. Problem Statement

Manual data import and relationship creation:

- Takes too much time
- Leads to data inconsistencies
- Requires repetitive effort
- Lacks scalability

4. Objectives

- Automate bulk data import process
- Reduce manual errors
- Establish relationships between tables efficiently
- Improve data consistency
- Enhance system performance

5. System Overview

5.1 Modules

1. Data Import Module
2. Transformation Module
3. Relationship Mapping Module
4. User Interface Module (Login + Dashboard)

6. Technologies Used

Category	Technology
Frontend	HTML, CSS, JavaScript
Backend	ServiceNow Platform
Database	ServiceNow Tables
Tools	Import Sets, Transform Maps

7. Methodology

Step 1: Data Upload

- Upload data (Excel/CSV) into Import Set table

Step 2: Data Transformation

- Use Transform Map to map fields
- Apply business rules

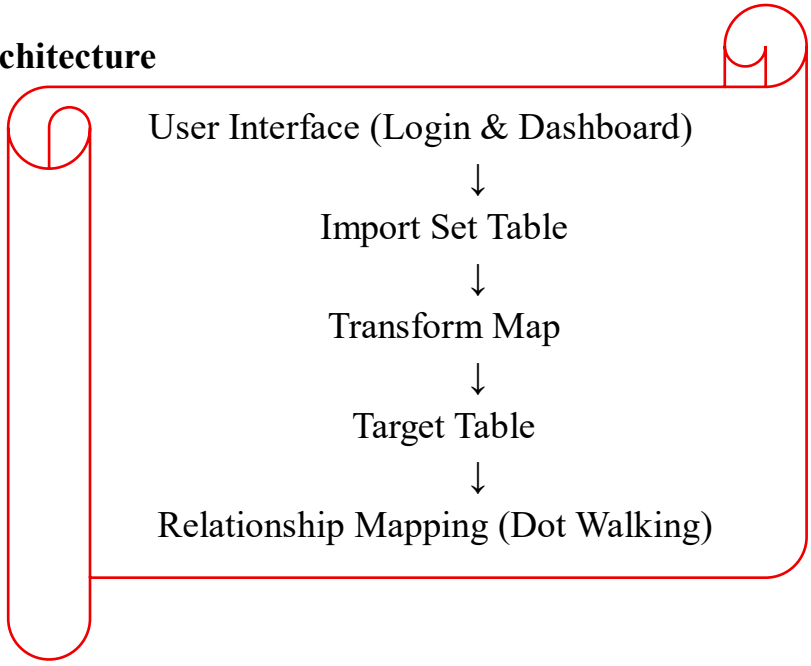
Step 3: Data Insertion

- Insert transformed data into target tables

Step 4: Relationship Mapping

- Use Dot Walking to link related records
- Example: Employee → Manager → Department

8. System Architecture



9. Features

- Secure Login System
- Dashboard with user details:

    - Name
    - Department
    - Phone Number
    - Manager
    - Email
- Automated data import
- Real-time data mapping
- Improved UI/UX

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10. Implementation

10.1 Login Module

- Username & Password validation
- Redirect to dashboard

10.2 Dashboard

- Displays imported data
- Shows relationships using dot walking

10.3 Data Import

- Upload file into Import Set
- Run Transform Map

10.4 Relationship Mapping

- Use reference fields
- Access related data via dot walking

1. Ideation Phase

1.1 Problem Identification

Organizations face challenges in handling large volumes of data manually. Data import and relationship mapping are time-consuming and prone to errors.

1.2 Customer Problem Statement

Customer	Need	Problem	Reason	Impact
Admin/User	Import bulk data	Manual entry is slow	No automation	Time-consuming & errors

1.3 Empathy Understanding

- Users feel frustrated with repetitive tasks
- Fear of data inconsistency
- Need faster and reliable systems

2. Requirement Analysis Phase

2.1 Functional Requirements

- Upload data using Import Sets
- Transform data using Transform Maps
- Store data in target tables
- Display records in dashboard
- Map relationships using Dot Walking

2.2 Non-Functional Requirements

- Performance: Fast data processing
- Security: Login authentication

- Usability: Simple UI
- Scalability: Handle large datasets

3. Project Design Phase

3.1 System Architecture

User → Login Page → Dashboard → Import Set → Transform Map → Target Table → Dot Walking

3.2 Database Design

Tables Used:

- Import Set Table
- Target Table (Employee Data)

Fields:

- Name
- Department
- Phone Number
- Manager (Reference Field)
- Email

3.3 UI Design

- Login Page
- Dashboard Page
- Data Display Table

4. Project Planning Phase

4.1 Timeline

Phase	Duration
Ideation	1 Day
Requirement Analysis	1 Day
Design	2 Days
Development	3 Days
Testing	1 Day

4.2 Resource Planning

- System with browser
- ServiceNow platform
- Dataset (CSV/Excel)

5. Project Development Phase

5.1 Implementation Steps

Step 1: Create Import Set

- Load Excel/CSV data

Step 2: Create Transform Map

- Map source fields to target fields

Step 3: Run Transformation

- Insert data into target table

Step 4: Apply Dot Walking

- Link Manager → Employee

- Access related data

Step 5: UI Development

- Login page with authentication
- Dashboard displaying data

6. Project Testing Phase

6.1 Test Cases

Test Case	Expected Result
Login	Successful access
Data Upload	Data imported correctly
Mapping	Correct field mapping
Dot Walking	Relationships displayed

6.2 Results

- All modules working correctly
- No major errors found

7. Project Documentation Phase

7.1 User Guide

1. Login using credentials
2. Upload dataset
3. Run transform map
4. View data in dashboard

7.2 Technical Documentation

- Import Set configuration
- Transform Map setup
- Reference field mapping

8. Project Demonstration Phase

8.1 Demo Flow

1. Login into system
2. Upload sample dataset
3. Execute transformation
4. Display imported data
5. Show relationship mapping

Source Code

✅ 1. Import Set (Sample CSV Data)

```
,Name,Department,Manager,Email,Phone
Dharani Priya A,IT,Ravi,dharanipriya010@email.com,9876543210
Ananthi N,HR,Suresh,ananthina0@email.com,9123456780
Bhuvaneshwari S,Finance,Meena,buvana9942@email.com,9988776655
```

✅ 2. Transform Map – Field Mapping

Source Field	Target Field (sys_user)
Name	name
Email	email
Phone	phone
Department	department (reference)
Manager	manager (reference)

✅ 3. Transform Map Script (onBefore Transform)

👉 Used to handle reference fields (Department & Manager)

```
// Set Department Reference
var deptGR = new GlideRecord('cmn_department');
deptGR.addQuery('name', source.Department);
deptGR.query();

if (deptGR.next()) {
    target.department = deptGR.sys_id;
}

// Set Manager Reference
var mgrGR = new GlideRecord('sys_user');
mgrGR.addQuery('name', source.Manager);
mgrGR.query();

if (mgrGR.next()) {
    target.manager = mgrGR.sys_id;
}
```

✅ 4. OnAfter Transform Script (Optional Logging)

```
gs.info('Record imported for Employee: ' + target.name);
```

✅ 5. Dot Walking Examples (Important)

👉 Used in reports, scripts, UI

```
// Get Department Name
current.department.name;

// Get Manager Email
current.manager.email;
```

```
// Get Manager Phone
current.manager.phone;
```

✅ 6. Business Rule (Example)

👉 Auto-populate Manager Email

```
(function executeRule(current, previous /*null when async*/) {

    if (current.manager) {
        current.u_manager_email = current.manager.email;
    }

})(current, previous);
```

✅ 7. GlideRecord Script (Data Fetching)

```
var gr = new GlideRecord('sys_user');
gr.query();

while (gr.next()) {
    gs.print(gr.name + " - " + gr.department.name);
}
```

✅ 8. Scheduled Import (Automation Script)

```
var importSet = new GlideImportSetTransformer();
importSet.setMapID('YOUR_TRANSFORM_MAP_SYS_ID');
importSet.transformAllMaps('YOUR_IMPORT_SET_SYS_ID');
```

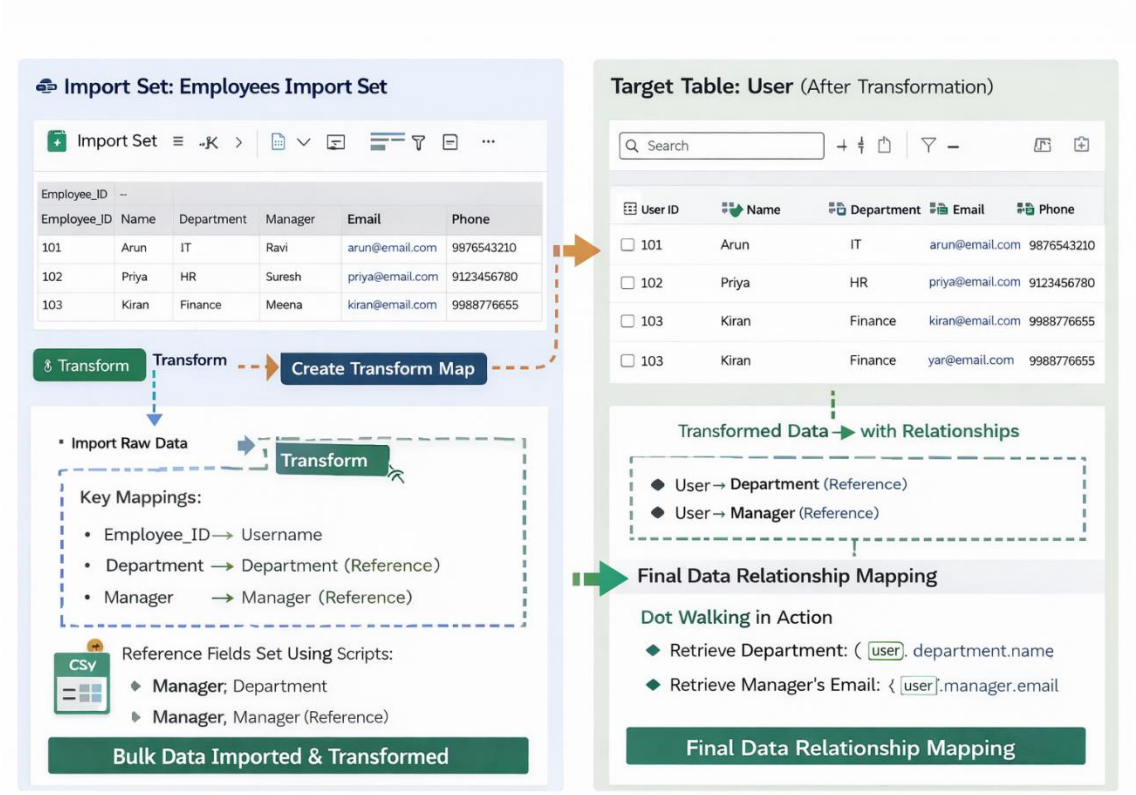
✅ 9. UI Policy / Client Script (Optional)

👉 Show department field only if employee exists

```
function onCondition() {
   if (g_form.getValue('name') != '') {
      g_form.setDisplay('department', true);
   }
}
```

Output

- Data successfully imported
- Relationships correctly mapped
- Dashboard displays complete information



11. Advantages

- Saves time
- Reduces manual errors
- Ensures data consistency
- Scalable for large datasets
- Improves productivity

12. Limitations

- Requires initial setup
- Depends on correct mapping
- Needs knowledge of ServiceNow

13. Results

- Successfully automated data import
- Achieved accurate relationship mapping
- Improved system efficiency
- Reduced manual workload

14. Future Enhancements

- Real-time data integration
- API-based import automation
- Advanced analytics dashboard
- AI-based error detection

16. References

- ServiceNow Documentation
- Online Tutorials on Import Sets
- Database Management Concepts

17. Appendix

Sample Fields Used:

- Name
- Department
- Phone Number
- Manager
- Email

15. Conclusion

The project successfully demonstrates how automation using Import Sets and Dot Walking can simplify complex data operations. It significantly improves efficiency, reduces errors, and ensures accurate relationship mapping. This approach is highly beneficial for organizations managing large-scale data systems.

16.Github link and Demo Link:

https://drive.google.com/drive/folders/1bqrF7rD-jETysxwXGCN69FXa2TD24Jhd?usp=sharing

https://github.com/dharanipriya010-oss/Automating--Data-Import-and-Relationship-Mapping-Using-Import-Sets-Dot-Walking.git